$$\phi(x) = \begin{bmatrix} 1 \\ x_1 \\ x_2 \\ \\ x_1^2 \\ x_1 x_2 \\ x_1 x_3 \\ \\ x_2 x_1 \\ \\ x^3 \\ x_1^2 x_2 \end{bmatrix}$$

monomial of x of degree ≤ 3

$\approx O(d^3)$

10^9

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \text{ not targets} \quad z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \quad x_4 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\phi(x) = \begin{bmatrix} x_1 x_1 \\ x_1 x_2 \\ x_1 x_3 \\ x_2 x_1 \\ x_2 x_2 \\ x_2 x_3 \\ x_3 x_1 \\ x_3 x_2 \\ x_3 x_3 \end{bmatrix}_{9\times 1} \quad \phi(z) = \begin{bmatrix} z_1 z_1 \\ z_1 z_2 \\ \vdots \\ z_3 z_3 \end{bmatrix}_{9\times 1} \quad \phi(x) = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 4 \\ 6 \\ 3 \\ 6 \\ 9 \end{bmatrix}_{9\times 1}$$

feature map

$$\phi(z) = \begin{bmatrix} 16 \\ 20 \\ 24 \\ 20 \\ 25 \\ 30 \\ 24 \\ 30 \\ 36 \end{bmatrix}$$

$$\phi(x) \cdot \phi(z) = 1024 \Rightarrow 16 + 40 + 72 + \dots$$

$$\boxed{(\langle x, z \rangle)^2 = K(x, z)}$$

$$K(x,z) = (4 + 10 + 18)^2 = (x_1 z_1 + x_2 z_2 + x_3 z_3)^2$$

Kernel function $= 1024$

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2\times1} \quad z = \begin{bmatrix} 3 \\ 4 \end{bmatrix}_{2\times1} \quad \mathbb{R}^2$$

$$\phi(x) = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2}\, x_1 x_2 \\ \sqrt{2}\, x_1 \\ \sqrt{2}\, x_2 \\ 1 \end{bmatrix}_{6\times1} \quad \mathbb{R}^6$$

$$K(x,z) = (x \cdot z + 1)^2$$

dot product of x & z

$$\phi(x) = \begin{bmatrix} 1 \\ 4 \\ 2.82 \\ 1.41 \\ 2.82 \\ 1 \end{bmatrix} \qquad \phi(z) = \begin{bmatrix} 9 \\ 16 \\ 16.97 \\ 4.24 \\ 5.65 \\ 1 \end{bmatrix}$$

$$\phi(x) \cdot \phi(z) = 143.76$$

$$K(x, z) = (x_1 z_1 + x_2 z_2 + 1)^2$$

$$= 144$$